EXECUTIVE SUMMARY

Diagnosing Revenue Concentration and Delivery Failures in a Global Distribution Network

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Strong overall financial results do not necessarily indicate a healthy business. Behind DataCo's positive overall performance are important operational inefficiencies and commercial dependencies that are not immediately visible in aggregate metrics.

This analysis investigates where revenue is concentrated, why a meaningful share of transactions generate losses, how logistics performance affects operations, and which areas leadership should prioritize to improve long-term commercial performance.

Business Objective

The objective of this project was to identify the commercial and operational factors influencing DataCo's performance and translate analytical findings into practical recommendations for improving profitability, logistics performance, and resource allocation.

Methodology

The project followed a structured business intelligence workflow:

Business Understanding → Data Audit → Data Preparation → Data Modeling → Exploratory Analysis → Business Insights → Recommendations → Power BI Dashboard

Using SQL Server, data quality was validated and exploratory analysis was performed on 180,519 order-line records. A star-schema data model and Power BI dashboard were then developed to support interactive business reporting.

Key Findings

- ### Delivery performance is the highest operational priority

More than 54% of orders were delivered late, making logistics performance the most significant operational issue identified. Standard Class shipping accounts for approximately 60% of shipments, making it the logical starting point for further investigation.

- ### Hidden losses reduce overall profitability

Although the company maintains a 10.78% overall profit margin, 21.15% of individual orders generate losses. This suggests that aggregate performance conceals a substantial number of underperforming transactions requiring root-cause analysis.

- ### Revenue concentration increases business risk

The Fan Shop department contributes approximately 47% of company sales and profit, while the Fishing category contributes approximately 19%, indicating a significant dependence on a limited number of revenue sources.

- ### Current discount strategy appears effective

Low and medium discount levels account for approximately 83% of sales and 84% of profit, suggesting that current pricing practices generally support profitability. However, additional analysis of highly discounted orders is required before changing pricing policy.

- ### Europe leads commercial performance

Europe generates the highest sales and profit among the five analyzed markets. Additional benchmarking is recommended before making region-specific investment decisions.

Priority Recommendations

1. Improve delivery performance by identifying the shipping methods and markets responsible for the highest delay rates.

2. Investigate loss-making transactions by analyzing shipping costs, discounts, destinations, and product mix.
3. Reduce dependency on Fan Shop and Fishing by diversifying revenue across additional departments and categories.
4. Maintain disciplined discount policies until high-discount profitability is fully evaluated.
5. Quantify market performance differences before increasing regional investment.

Business Impact

The analysis indicates that DataCo's strongest opportunities lie in operational improvement rather than revenue generation alone. Addressing delivery delays, reducing loss-making transactions, and mitigating revenue concentration would strengthen profitability while reducing commercial risk.

At a Glance

Metric	Result
Sales (2015–2018)	$36.78M
Profit Margin	10.78%
Late Deliveries	54.8%
Loss-making Orders	21.15%
Largest Revenue Driver	Fan Shop (~47%)
Markets	5
Countries	164
Orders	65,752
Customers	20,652
Tools	SQL Server, Power BI

Full analysis, SQL scripts, Power BI dashboard, and documentation:
https://github.com/FARAHBENSALEM01/DataCo-Commercial-Distribution-Analysis/tree/main